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AI-Powered Resume & Interview Preparation Platform



A smart career-preparation assistant for resume improvement, personalized mock interviews, and actionable feedback.

The platform combines:- Resume analysis and improvement- Job-skill matching- AI-based mock interviews- Personalized feedback- Performance tracking It provides users with a structured and personalized approach to interview preparation.

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Problem Statement

Why do students need a smarter way to prepare for interviews?

Resume gaps

Candidates may struggle to present skills, projects, and achievements clearly.

Limited practice

Access to realistic technical and HR mock interviews can be limited.

Unclear feedback

Candidates often need specific guidance on what to improve next.

Project goal

Create one platform that connects resume analysis, job-skill matching, personalized interview practice, and progress tracking—so users can prepare with a clearer plan.

Project Objectives

A practical, student-focused career preparation workflow

01

Analyze resumes

Extract education, skills, projects, and experience from uploaded files.

02

Improve job alignment

Compare a resume with a target job description and highlight possible skill gaps.

03

Practice interviews

Generate relevant technical and HR questions based on the user's profile.

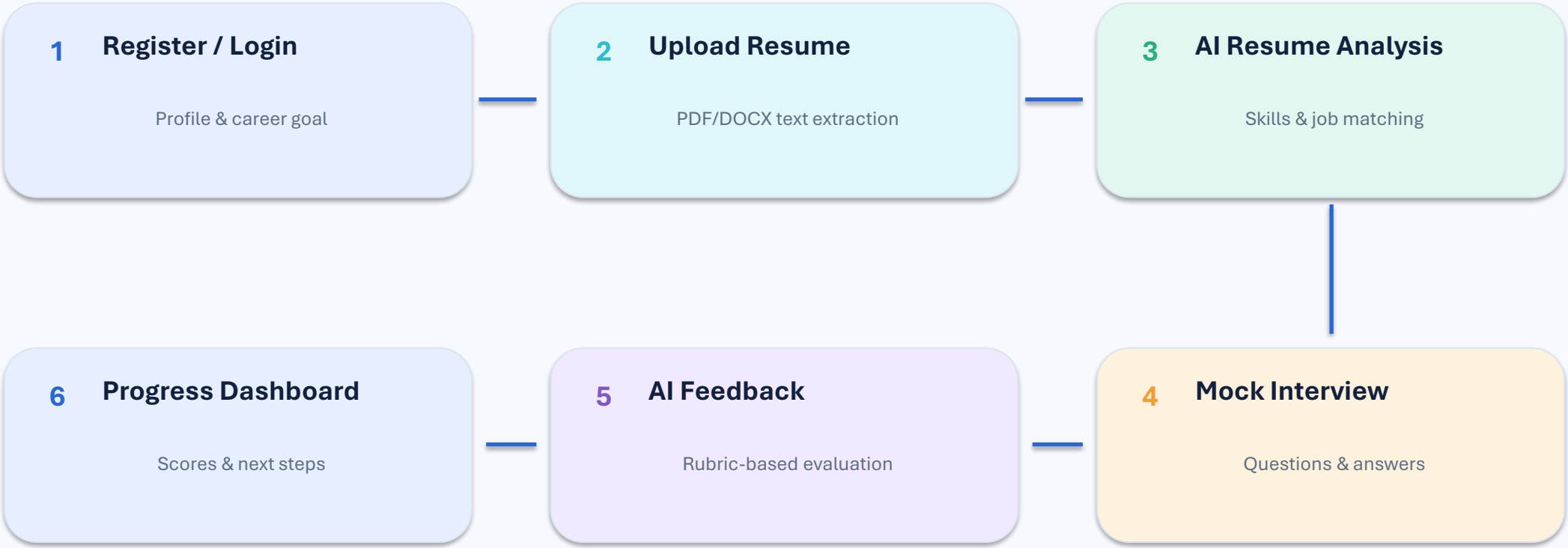
04

Give useful feedback

Evaluate answers with a clear rubric and suggest next steps.

Workflow

From user input to personalized improvement plan



Output: a clearer resume, targeted interview practice, and measurable learning progress.

Core Modules

The platform is divided into four connected modules

1. AI Resume Analyzer

Upload PDF/DOCX • Extract resume text • Detect skills and sections • Compare against a job description

2. AI Resume Builder

Enter profile details • Improve bullet points • Generate a summary • Export a formatted resume

3. AI Mock Interview

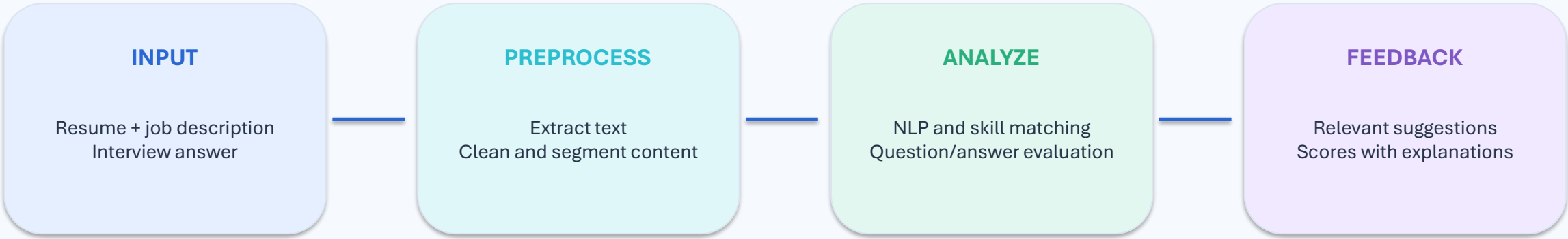
Select a role • Generate technical/HR questions • Submit typed or spoken answers • Receive feedback

4. Performance Dashboard

Track interview attempts • View feedback and scores • Identify strengths • Plan future practice

How AI Supports the Platform

An explainable pipeline instead of a single black-box score



Possible techniques

- PDF/DOCX text extraction
- TF-IDF or embeddings for resume–job matching
- Hugging Face models for selected NLP tasks

Evaluation principles

- Use a transparent answer rubric
- Explain why feedback was suggested
- Protect resumes and interview responses

Proposed System Architecture

USER INTERFACE	Web frontend • Resume upload • Interview room • Feedback dashboard
APPLICATION BACKEND	Authentication • File validation • API routes • Session management
AI / NLP SERVICES	Text extraction • Skill matching • Question generation • Answer evaluation
DATA & DEPLOYMENT	User/session database • File handling • Docker (if supported) • Logs

Technology Stack

Frontend

Existing frontend framework
HTML/CSS/JavaScript as needed

Backend

Python + Flask/FastAPI
Choose one compatible framework

NLP / AI

Hugging Face Transformers
TF-IDF or sentence embeddings

File processing

PyMuPDF for PDFs
python-docx for DOCX

Storage

SQLite for prototype
Protect user files and sessions

Deployment

Docker / Compose if configured
Environment variables for secrets

Implementation Roadmap

1

PHASE 1

Run the repository

Install dependencies; understand frontend/backend; fix setup errors.

2

PHASE 2

Resume analyzer

Upload PDF/DOCX; extract text; show detected sections and skills.

3

PHASE 3

Resume feedback

Add job description matching and practical suggestions.

4

PHASE 4

Mock interview

Generate questions; collect answers; evaluate with a clear rubric.

5

PHASE 5

Dashboard & testing

Save results; show progress; test privacy and edge cases.

Expected Benefits & Responsible Use

Benefits for candidates

- More structured preparation
- Clear resume improvement ideas
- Repeatable interview practice
- Progress tracking over time

Responsible AI considerations

- Ask permission before storing data
- Avoid exposing personal information
- Explain scores and limitations
- Let users review AI-generated content

CONCLUSION

project aims to combine resume analysis, personalized mock interviews, and progress tracking in one accessible experience. The first milestone is a reliable working prototype; advanced AI features can be added after the core workflow is tested.

Thank you

